CSE-251

Electronic Circuits

Lecture-15

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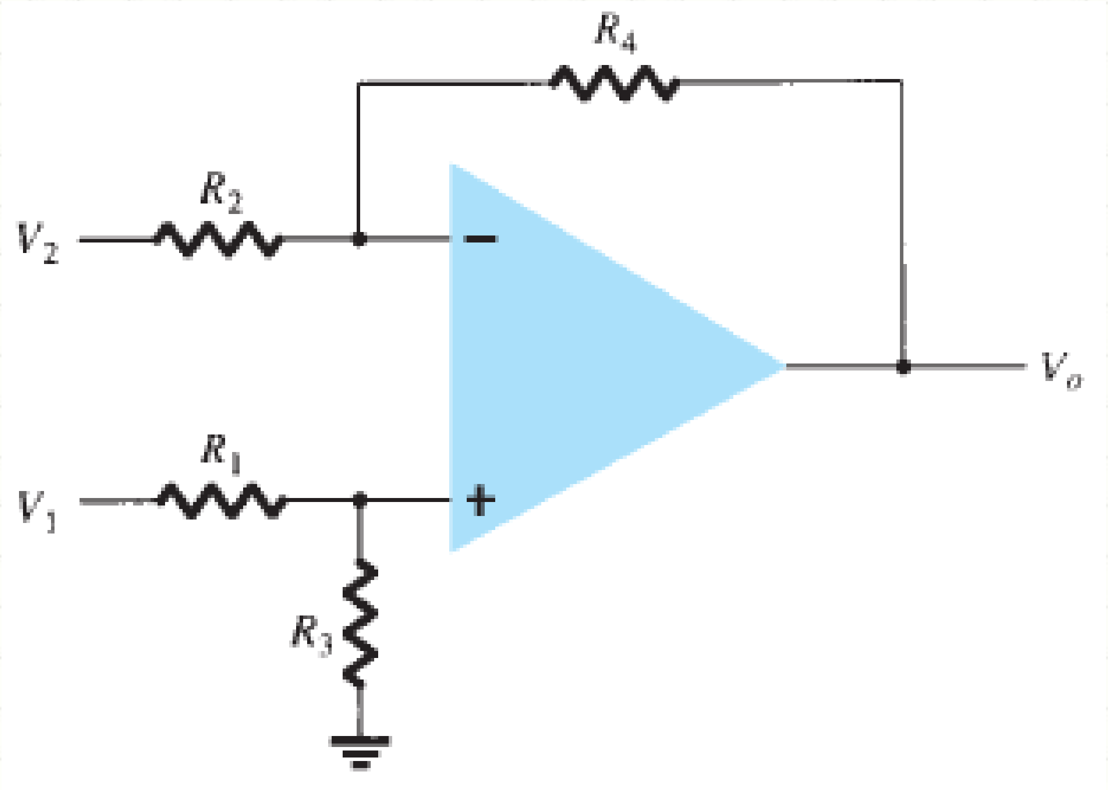
Today's Overview:

- ❑*Subtracting Amplifier*
- ❑*Voltage Follower*
- ❑*Differentiator*
- ❑*Integrator*

Subtracting Amplifier

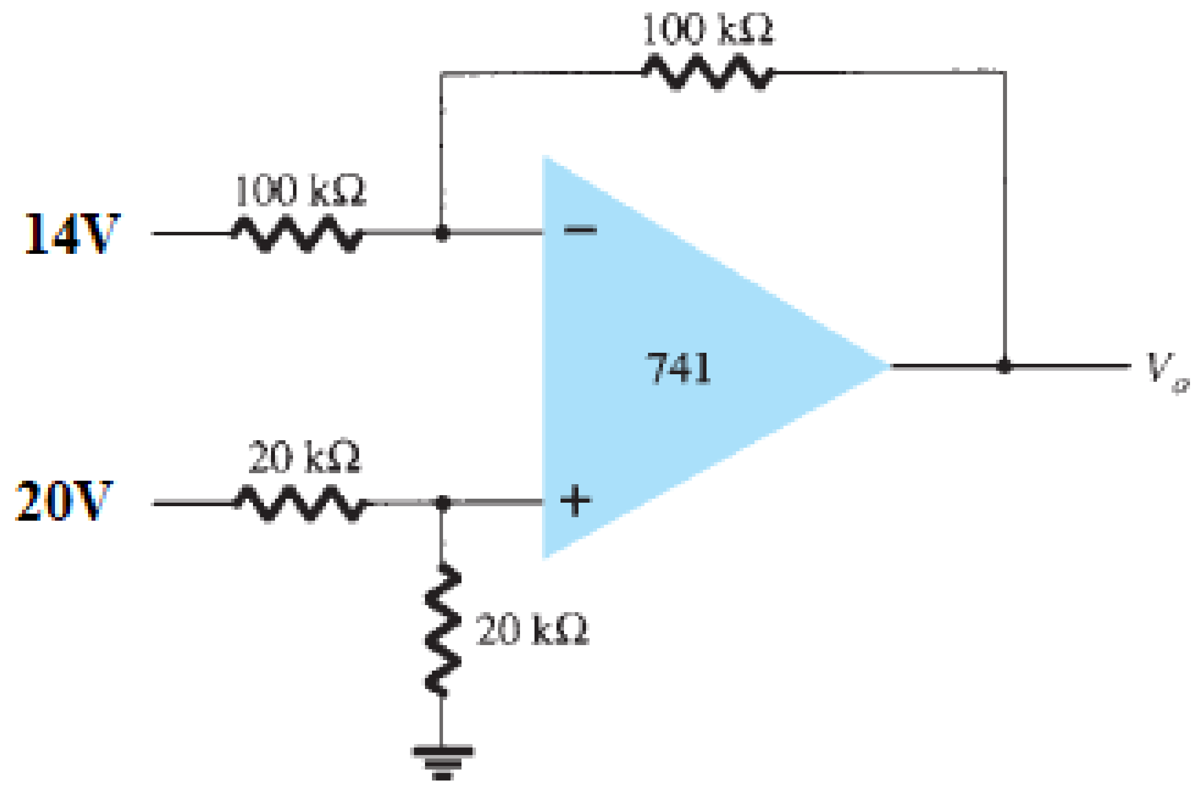
Subtracting/Difference Amplifier:



$$V_o = \frac{R_3}{R_1 + R_3}\frac{R_2 + R_4}{R_2}V_1 - \frac{R_4}{R_2}V_2$$

Problem:

EXAMPLE 11.8 Determine the output voltage for the circuit of Fig. 11.12.



Problem:

Solution: The resulting output voltage can be expressed as

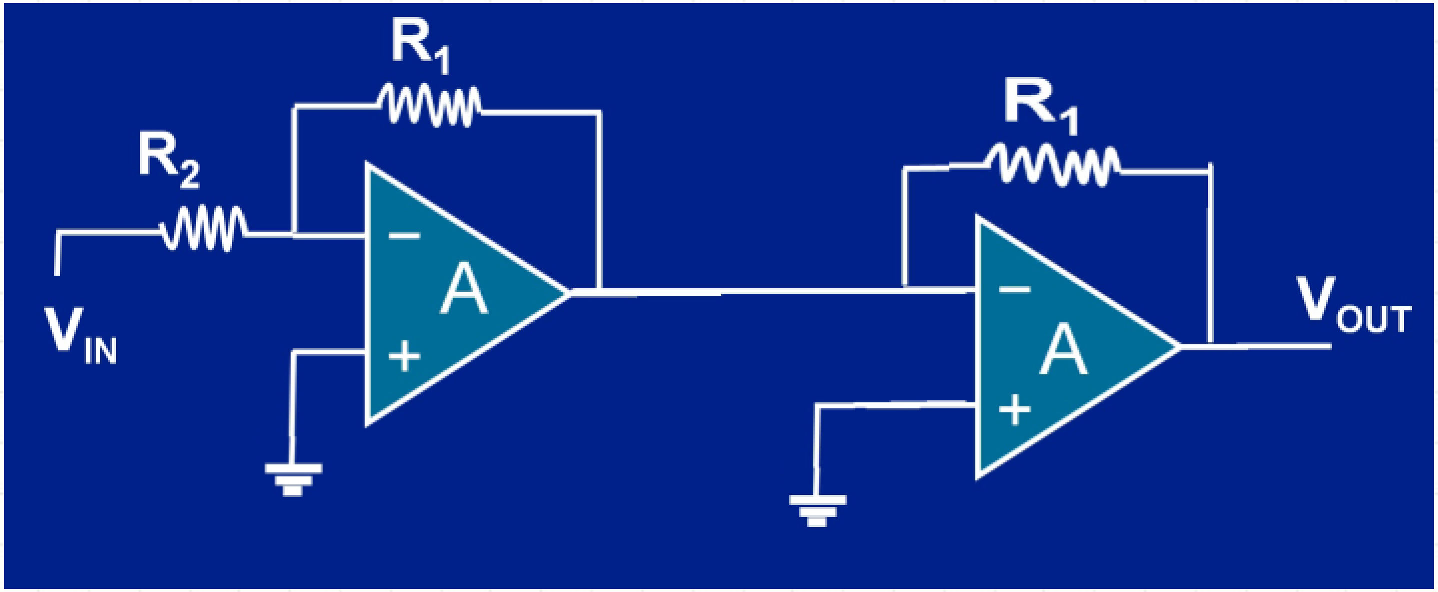
$$V_o = \left(\frac{20\,\text{k}\Omega}{20\,\text{k}\Omega + 20\,\text{k}\Omega}\right)\left(\frac{100\,\text{k}\Omega + 100\,\text{k}\Omega}{100\,\text{k}\Omega}\right)V_1 - \frac{100\,\text{k}\Omega}{100\,\text{k}\Omega}V_2$$

$$= V_1 - V_2 = 20 - 14 = 6V$$

Voltage Follower

Voltage Follower/Voltage Buffer:

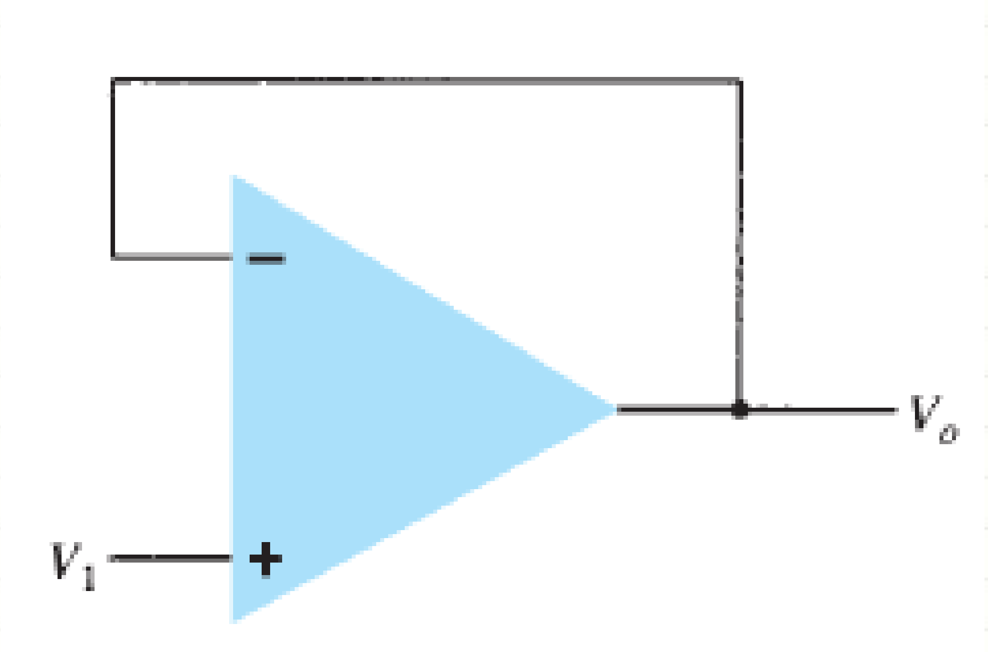
- ***Using inverting amplifier:***



$$\boxed{V_o = V_1}$$

Voltage Follower/Voltage Buffer:

- ***Using non-inverting amplifier:***



$$\boxed{V_o = V_1}$$

Differentiator

Differentiator Op-amp:

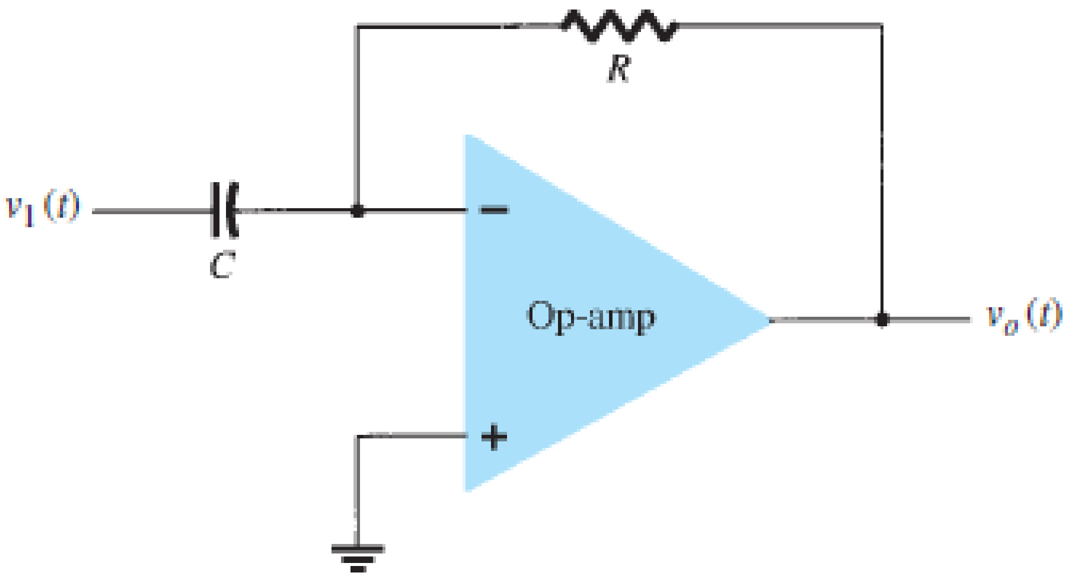
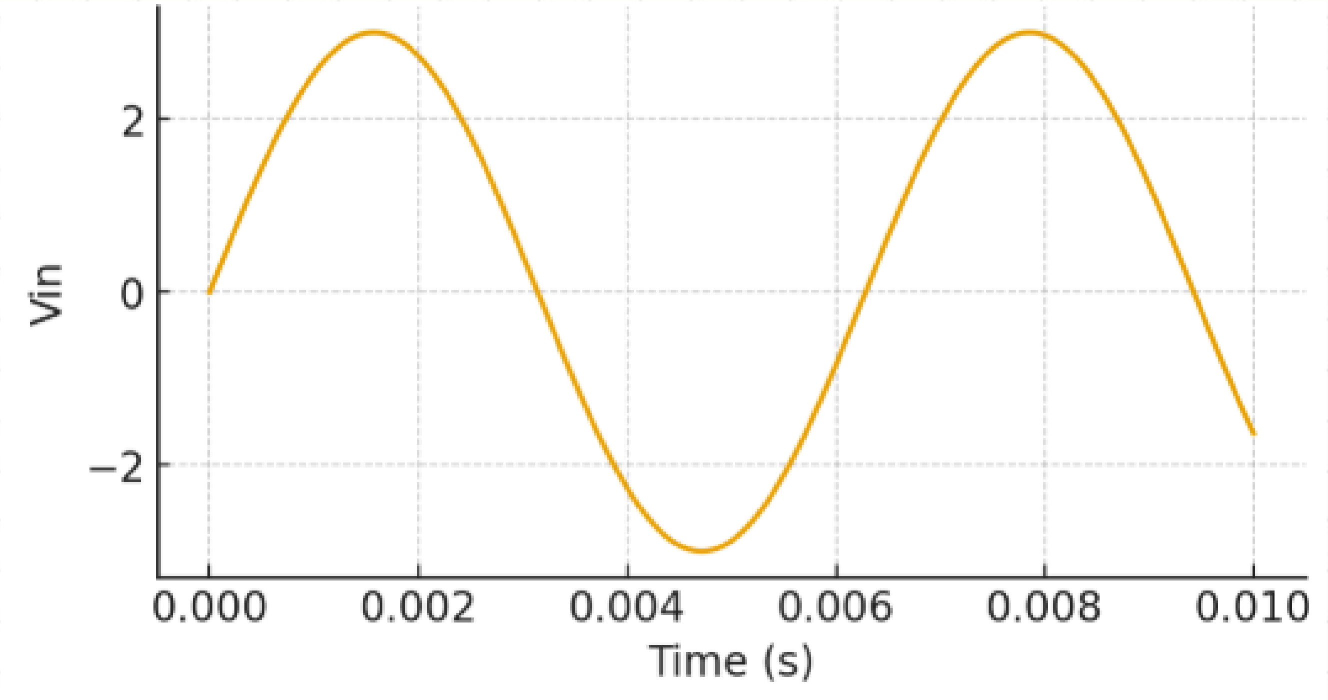


FIG. 10.41
Differentiator circuit.

$$v_o(t) = -RC\frac{dv_1(t)}{dt}$$

Problem:

Find $v_o(t)$ for $v_{in} = 3\sin(1000t)$, $R=5k\Omega$, $C=0.02\mu F$.



Problem:

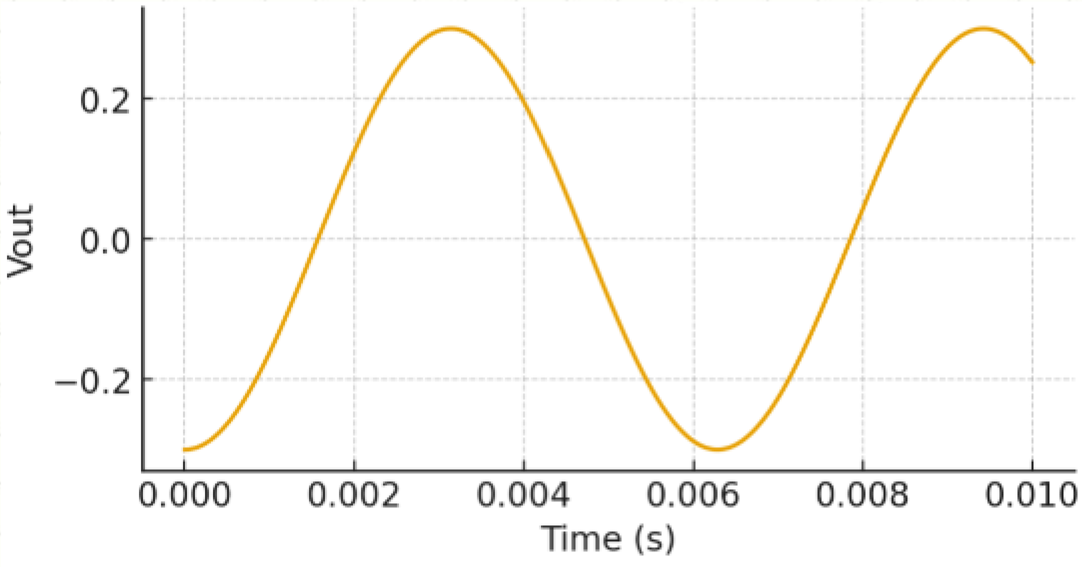
Solve:

$$\frac{d}{dt}[3\sin(1000t)] = 3000\cos(1000t)$$

$$RC = 5 \times 10^3 \times 2 \times 10^{-8} = 10^{-4}$$

$$v_o(t) = -10^{-4}(3000\cos(1000t))$$

$$v_o(t) = -0.3\cos(1000t)$$



Problem:

The input is a triangular wave oscillating from 0V to 10V with a period of 2ms, $R=100k\Omega$, $C=0.1\mu F$. Find full period output of the differentiator, showing the relationship between the input and output.

Problem:

Solve:

1. **Rising Edge:**

   The slope is:

$$\frac{dV_{in}}{dt} = \frac{10V - 0V}{2\,\text{ms}} = 5\,\text{V}/\text{ms}$$

   Using the differentiator equation:

$$v_o(t) = -RC\frac{dV_{in}}{dt}$$

$$v_o(t) = -100 \times 10^3 \times 0.1 \times 10^{-6} \times 5 = -0.05\,\text{V}/\text{ms}$$

2. **Falling Edge:**

   The slope is:

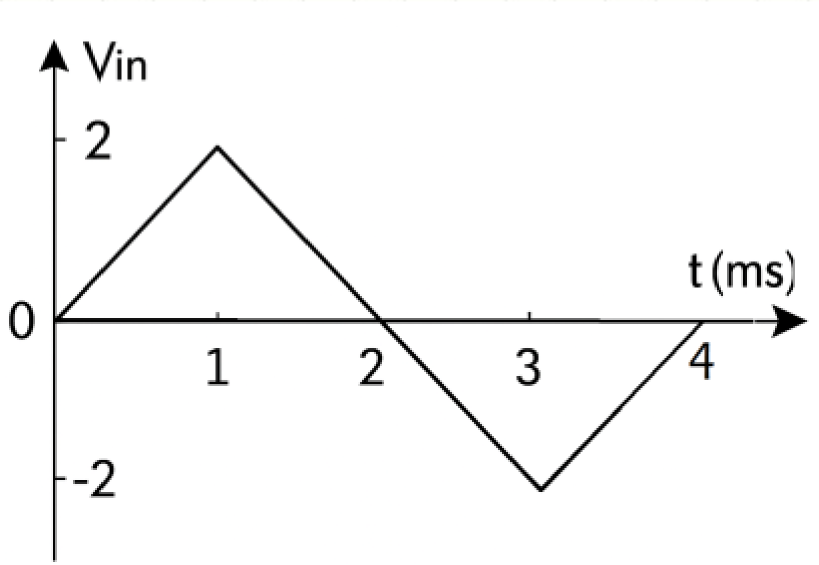
$$\frac{dV_{in}}{dt} = \frac{0V - 10V}{2\,\text{ms}} = -5\,\text{V}/\text{ms}$$

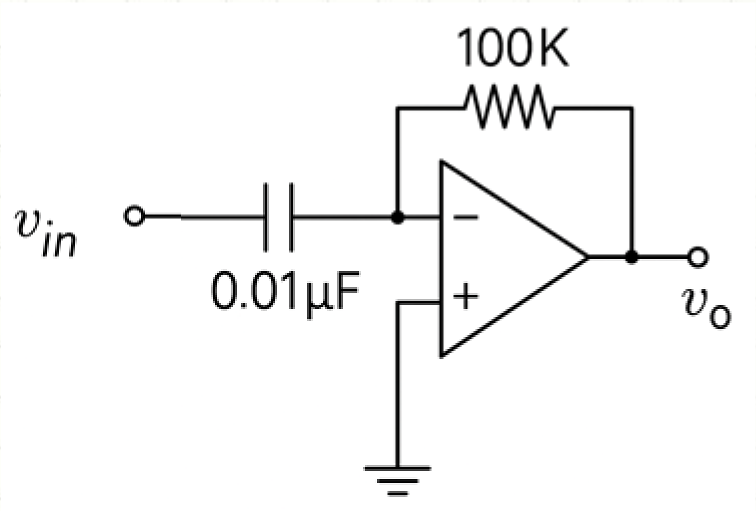
   Using the same differentiator equation:

$$v_o(t) = -100 \times 10^3 \times 0.1 \times 10^{-6} \times (-5) = 0.05\,\text{V}/\text{ms}$$

Problem:

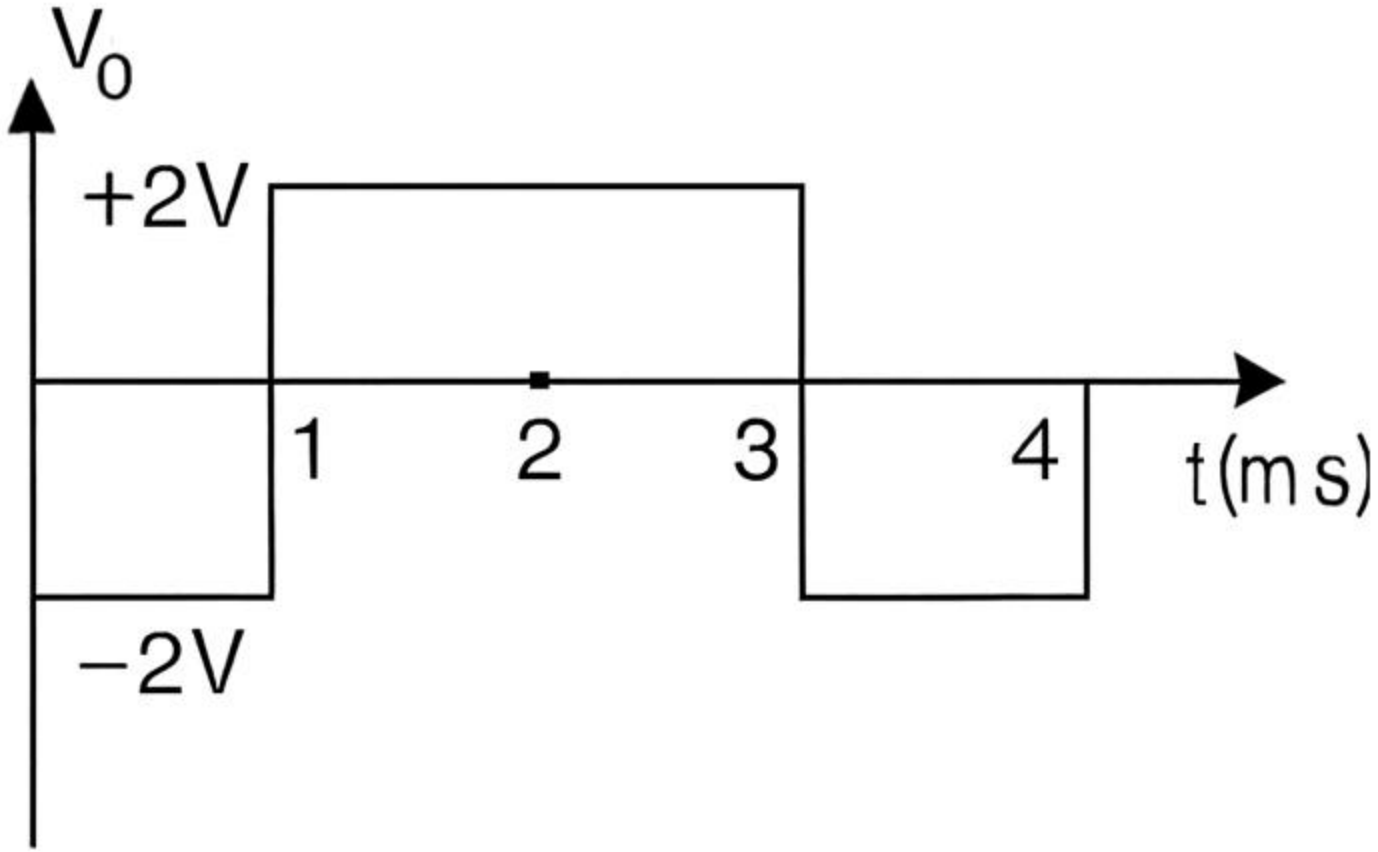
Plot the output waveform of the given differentiator op-amp.





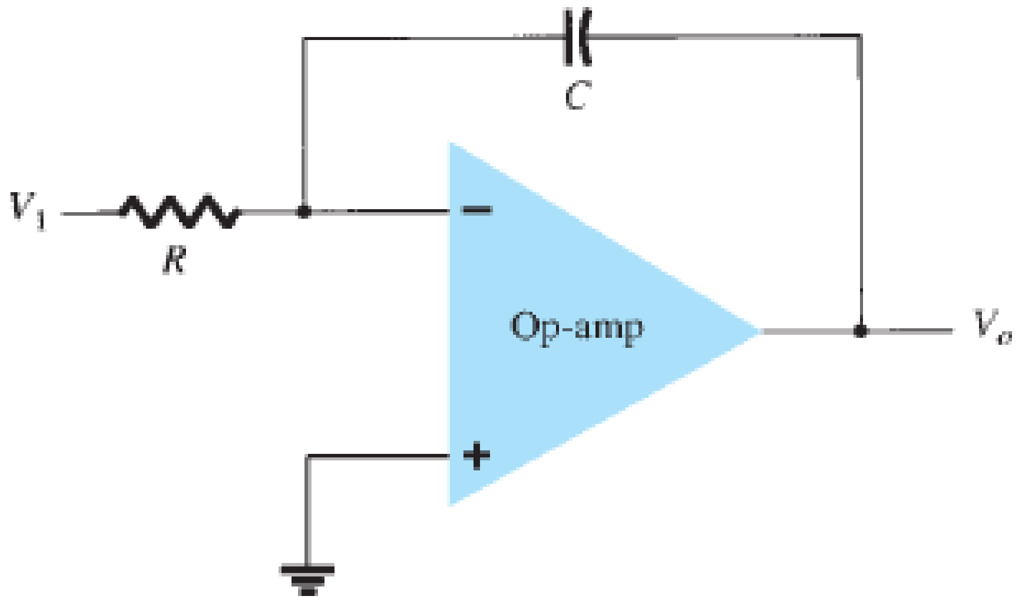
Problem:

Solve:



Integrator

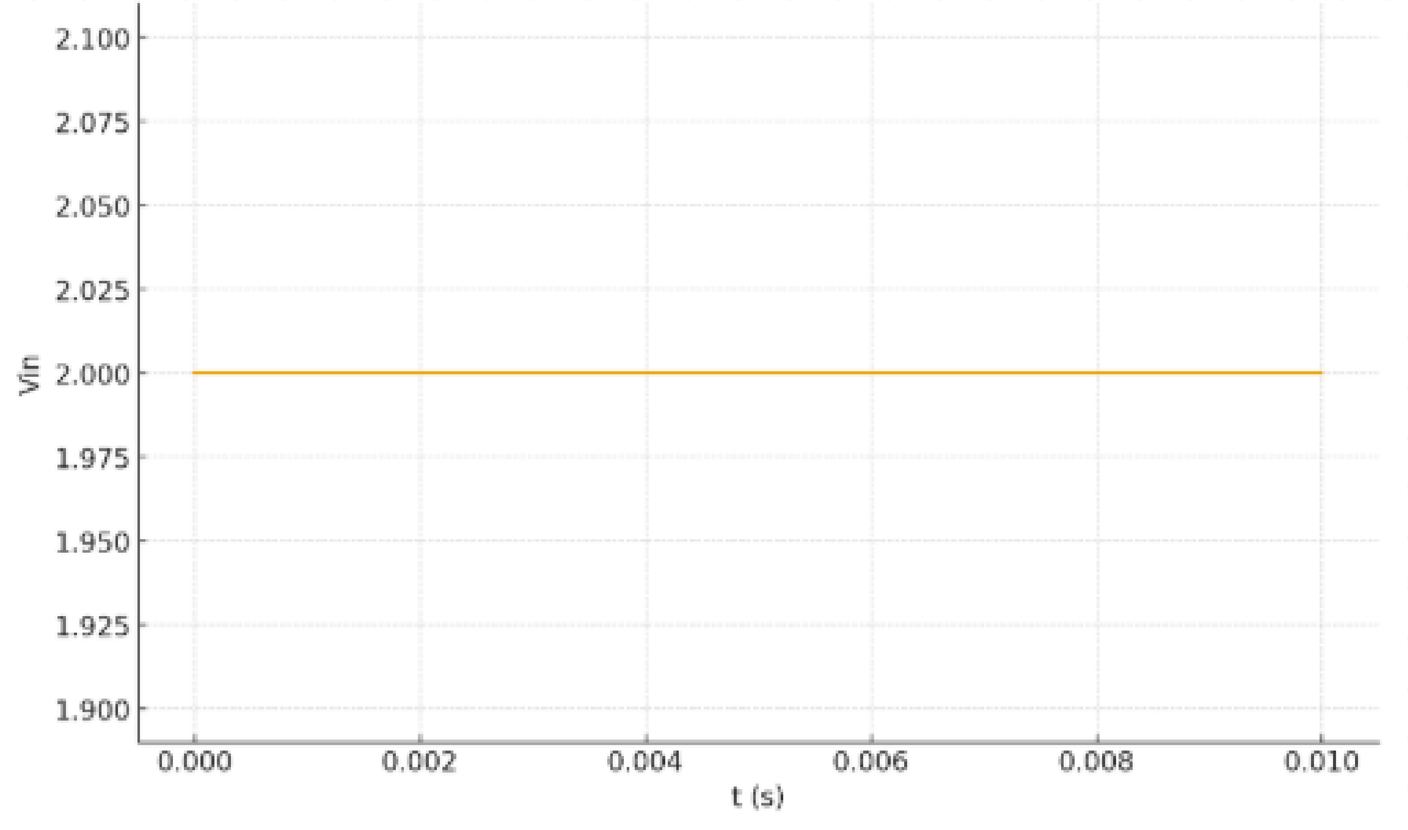
Integrator Op-amp:



$$\boxed{v_o(t) = -\frac{1}{RC}\int v_1(t)\,dt}$$

Problem:

An op-amp integrator has R=100 kΩ, C=0.1 μF. If the input is a constant voltage: v_{in} = 2V, then find the output voltage expression $v_o(t)$.



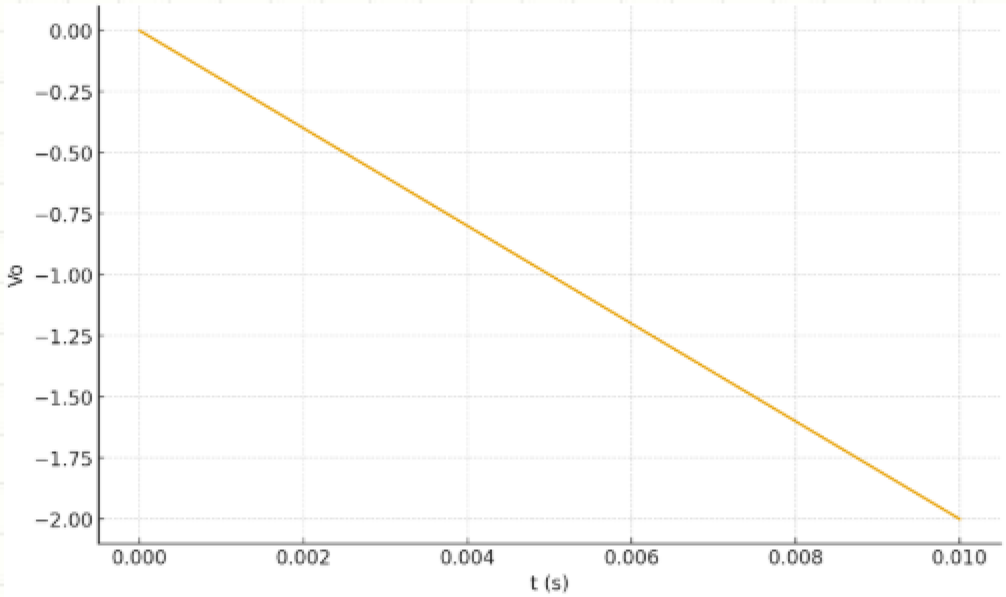
Problem:

Solve:

$$v_o(t) = -\frac{1}{RC}\int v_{in}dt$$

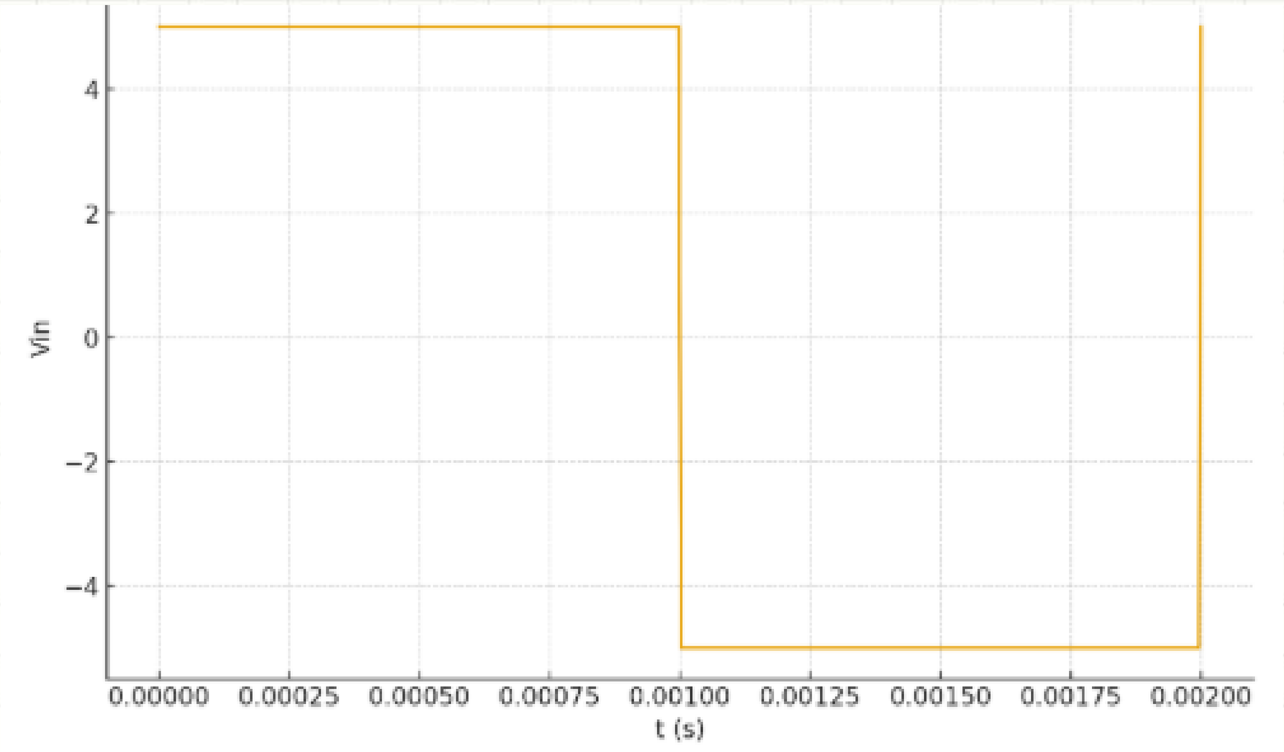
$$v_o(t) = -\frac{1}{100 \times 10^3 \cdot 0.1 \times 10^{-6}}(2t)$$

$$v_o(t) = -200t$$



Problem:

A square wave of amplitude 5V (switching between +5V and –5V) is applied to an integrator with R=10kΩ, C=0.01μF. Find the slope of $v_o(t)$ in each half cycle.



Problem:

Solve:

Slope of output:

$$\frac{dv_o}{dt} = -\frac{1}{RC} v_{in}$$

$$RC = 10^4 \cdot 10^{-8} = 10^{-4}$$

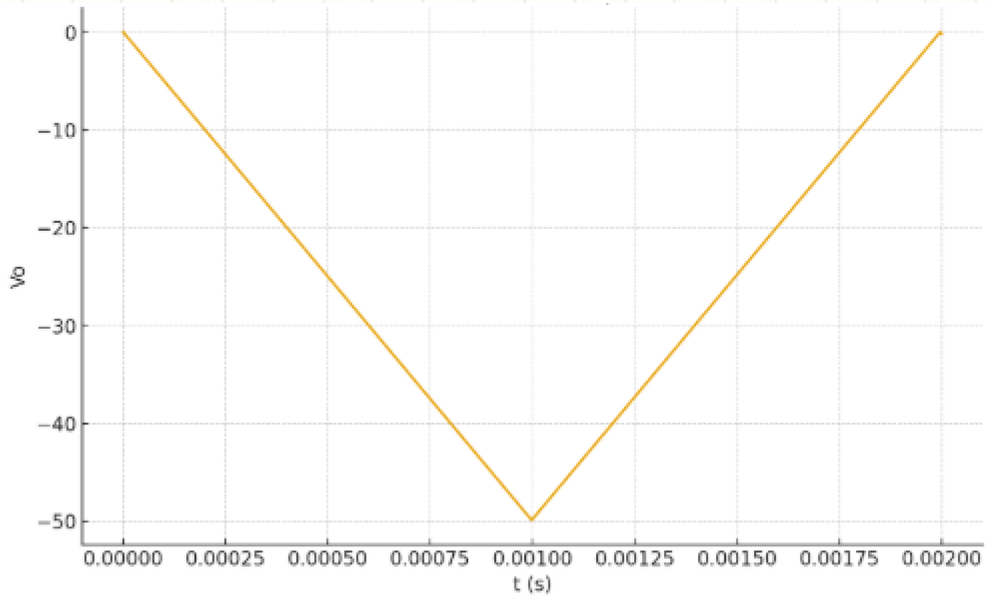
Thus:

- When $v_{in} = +5$:

$$\frac{dv_o}{dt} = -\frac{5}{10^{-4}} = -5 \times 10^4 = -50,000\, V/s$$

- When $v_{in} = -5$:

$$\frac{dv_o}{dt} = +50,000\, V/s$$



Problem:

Find $v_o(t)$ for $v_{in} = 2\sin(1000t)$ and $RC=1$.

Problem:

Solve:

$$v_o(t) = -\frac{1}{RC}\int 2\sin(1000t)\,dt$$

$$= -2\left(\frac{-\cos(1000t)}{1000}\right)$$

$$= \frac{2}{1000}\cos(1000t)$$

$$= 0.002\cos(1000t)$$

QUESTIONS?

See You In The Next Class